

9.8 Conclusion

The use of LLMs increases your flexibility to a large extent: You can choose between commercial models provided by providers such as OpenAI or Anthropic. Depending on the contract model, these providers guarantee that your data is protected against unauthorized access.

If you don't trust this guarantee or want to experiment with other language models, you can run free language models locally on your PC, on a central



notebooks is insufficient for the LLM version. Regardless of where the language model is executed, you can use it as a chatbot, via plug-in as a code assistant, or via API in your own applications.

In the open-source sector, you can choose from a wide range of models. Depending on the architecture and size, however, their implementation is very resource-intensive. The smaller versions, such as Llama 3.1-8B, can be run on more powerful desktop systems with a separate graphics card. For the larger models such as Llama 3.1-70B or even the 405B variant, however, you need very powerful hardware. Running such a large model is usually not worthwhile for individuals or small companies, as the infrastructure is disproportionately expensive. In this case, you return to one of the commercial providers.

In our tests, we've found that LLMs which run locally on private notebooks or PCs can't keep up with commercial cloud offerings. This is particularly true for the coding application: the token speed is too low, and the quality of the answers is noticeably lower. We expect this to change in the future: first, future computers will provide better hardware requirements for the execution of LLMs; and second, intensive research is currently being carried out to reduce the size of LLMs while maintaining the same quality.

